

QUESTION PAPER

Name of the Examination: FALL 2023-2024 – CAT-2

Course Code: CSE2013

Course Title: Information Theory and Coding

Set number: 2

Date of Exam: 16/10/2023 (Fri) (B1)

Duration: 90 min

Total Marks: 50

- Q1.** Consider a sequence of letters from English alphabets with their probability of occurrence as follows:

Letter	A	I	T	P	V
Probability	0.30	0.25	0.15	0.20	0.10

Construct the Huffman-tree and determine the source code for the word “**VITAP**”. Also, determine the % compression obtained with Huffman coding when the same word “**VITAP**” is encoded using ASCII. **(10M)**

- Q2.** Encode the message “**SCOPE**” where the characters have the following probabilities $S = 0.15$, $C = 0.1$, $O = 0.2$, $P = 0.2$ and $E = 0.35$. Implement arithmetic coding and determine the source code for the word “**CSE**”. **(10M)**

- Q3.** A source outputs two symbols A and B with probabilities $P(A) = 4 P(B)$. Apply Shannon-Fano source coding algorithm and determine the codes for A, B and evaluate the efficiency. **(10M)**

- Q4.** In a (5, 2) linear block code, the received codeword $[v]$ has five bits and are given as: $[r_1 k_1 r_2 k_2 r_3]$. Here, k_i represents the source code bits and r_i represents the redundant bits. It is known that r_1 and r_2 are the LSB and MSB of the source code, respectively whereas r_3 is always “1”.

(a) Prepare the list of all the codewords.

(b) Prove that the codewords in this logic are NOT valid codewords

(c) Propose a modification to this method so that the codewords will become valid. Note that while proposing the modification, you are strictly allowed to make only ONE modification to the logic behind any one redundant bit. It can either be r_1 , r_2 or r_3 .

(d) The 3rd and 4th bits of the codeword [i.e., v_3 and v_4] are corrupted during transmission. Verify whether the receiver can detect and as well as correct these two errors.

(e) Determine the efficiency of this method.

(5 x 2 = 10M)

- Q5.** Prepare a (n,k) Hamming Code such that the resulting codeword should be exactly 3 bits long. To solve this question, you can consider any k -bit source code of your choice that satisfies the Hamming Code condition. Now induce an error in the 2nd and 3rd bit positions of the codeword. Perform Hamming Code operation at the receiver and prove that Hamming Codes cannot correct multiple errors. (10M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	2	1,2	-	-	10
Q2	3	2	1,2	-	-	10
Q3	4	3	1,2	-	-	10
Q4	4	3	1,2	-	-	10
Q5	4	3	1,2	-	-	10